PHYS 135B Discussion 4

TA: Yifan Liu

yliu545@ucr.edu

I. Vector calculus identities (HW3 Q4)

(e) $\triangle(a\mathbf{b})$

We use the Leibniz rule repeatly in this case

$$\begin{aligned}
\triangle(a\mathbf{b}) = \nabla\cdot\nabla(a\mathbf{b}) &= \sum_i \partial_i^2(a\sum_j b_j\mathbf{e}_j) \\
&= \sum_{ij} \mathbf{e}_j\partial_i^2(ab_j) \\
&= \sum_{ij} \mathbf{e}_j\partial_i[(\partial_i a)b_j + a\partial_i b_j] \\
&= \sum_{ij} \mathbf{e}_j[\partial_i^2 ab_j + 2\partial_i a\partial_i b_j + a\partial_i^2 b_j] \\
&= \sum_i [\partial_i^2 a\mathbf{b} + 2\partial_i a\partial_i\mathbf{b} + a\partial_i^2\mathbf{b}]
\end{aligned}$$

Therefore,

$$\triangle(a\mathbf{b}) = a\triangle\mathbf{b} + 2\nabla a\cdot\nabla\mathbf{b} + (\triangle a)\mathbf{b}$$

II. Passive transformation (HW3 Q7)

There are two types of spacial transformations: active and passive transformation, in which we transform the physical position and the coordinate systerm, respectively. See wikipedia(`https://en.wikipedia.org/wiki/Active_and_passive_transformation`)

Since we only transform the coordinate system (passive transformation), the $\mathbf{B}$ field measured at P in the old coordinate system should be the same as the $\mathbf{B}'$ field measured

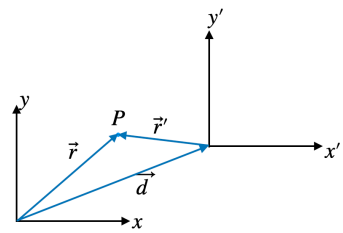

Figure 1: Passive transformation

in the new coordinate system at P, where the prime indicates that the magnetic field is measured in the new coordinate system. Therefore,

$$\mathbf{B}'(\mathbf{r}') = \mathbf{B}(\mathbf{r}) \rightarrow \mathbf{B}'(\mathbf{r}') = \mathbf{B}(\mathbf{r}' + \mathbf{d})$$

Notice that we only translate the coordinate system without changing the orientation of the axes, thus the unit vectors should be the same in the new coordinate system, i.e. $\mathbf{e}'_i = \mathbf{e}_i$ $(i = x, y, z)$. Thus, we obtain

$$B'_i(\mathbf{r}') = \mathbf{B}'(\mathbf{r}') \cdot \mathbf{e}'_i = \mathbf{B}(\mathbf{r}' + \mathbf{d}) \cdot \mathbf{e}_i = \boxed{B_i(\mathbf{r}' + \mathbf{d})}$$

$$\begin{aligned} B'_x(\mathbf{r}') &= 16(y' + 6) \\ B'_y(\mathbf{r}') &= 8[x' + 4 + (z' + 6)(y' + 6)] \\ B'_z(\mathbf{r}') &= 40 \end{aligned}$$

Note: if we rotate the coordinate system as well, then in general $\mathbf{e}'_i \neq \mathbf{e}_i$ for all i. Then we need to find the ratation matrix $\hat{R}$ which transforms $\mathbf{e}_i$ into $\mathbf{e}'_i$, i.e. $\mathbf{e}'_i = \hat{R}\mathbf{e}_i$.

Think: what are the components of **B** field in the new coordinate system if we shift the origin to (4,6,6) followed by a rotation of the axes around z by an angle θ?

III. Approximated B field at the center of a circular current loop (HW3 Q12)

To calculate the field a the center of a circular wire loop, we pproximate the circle through a regular polygon that encloses the circle (e.g. regular hexagon Fig.2). Based on discrete rotational symmetry, the contribution to the B field from each side should be the same.

Now let's calculate the B field produced by each side (Fig.3). Notice that the position of field point is $\mathbf{r} = R\mathbf{e}_y$, and current element is x away from the origin along the x-axis, i.e. $\mathbf{r}' = x\mathbf{e}_x$. Furthermore, we have

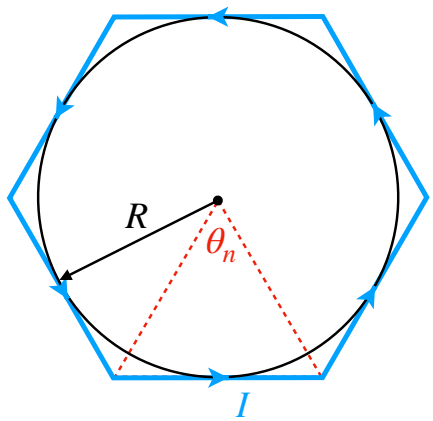

Figure 2: A circular loop approximated by a hexagon loop

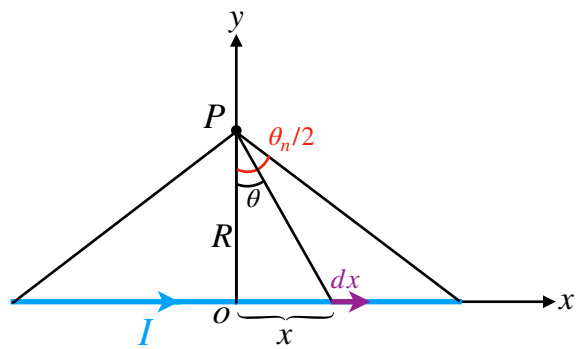

Figure 3: Contribution to the B field from each side

$$
\begin{aligned}
d\mathbf{r}' \times (\mathbf{r} - \mathbf{r}') &= dx\mathbf{e}_x \times (R\mathbf{e}_y - x\mathbf{e}_x) = Rdx\mathbf{e}_z \\
dx &= d(R\tan\theta) = R\cos^{-2}\theta d\theta \\
|\mathbf{r} - \mathbf{r}'| &= \sqrt{R^2 + x^2}
\end{aligned}
$$

Therefore, the magnetic field from each side $\mathbf{B}_n$ is

$$
\begin{aligned}
\mathbf{B}_n &= \int \frac{I}{c}\frac{d\mathbf{r}' \times (\mathbf{r} - \mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|^3} = \int \frac{I}{c}\frac{Rdx\mathbf{e}_z}{|R^2 + x^2|^{3/2}} \\
&= \int_{-\theta_n/2}^{\theta_n/2} \frac{I}{c}\frac{R^2\cos^{-2}\theta d\theta \mathbf{e}_z}{|R^2 + R^2\tan^2\theta|^{3/2}} \\
&= \int_{-\theta_n/2}^{\theta_n/2} \frac{I}{c}\frac{\cos\theta d\theta}{R}\mathbf{e}_z \\
&= \frac{2I\sin(\theta_n/2)}{cR}\mathbf{e}_z
\end{aligned}
$$

Note: For an infinite current wire, we let $\theta_n/2 \to \pi/2$, thus $B \to \frac{2I}{cR}$.

The total field at the center thus is

$$
\mathbf{B} = n\mathbf{B}_n = \frac{2I}{cR}n\sin(\theta_n/2)\mathbf{e}_z \xrightarrow{n\to\infty} \frac{2\pi I}{cR}\mathbf{e}_z
$$

IV. B field at the center of a circular current loop (HW3 Q13)

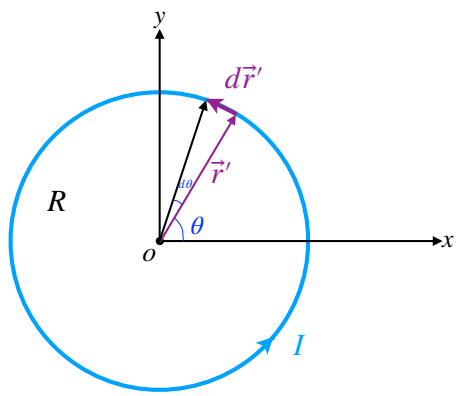

Figure 4: A circular wire loop

The magnetic field **B** is

$$\begin{aligned}\mathbf{B}(\mathbf{r}=\mathbf{0}) &= \int \frac{I}{c}\frac{d\mathbf{r}'\times(-\mathbf{r}')}{|\mathbf{r}'|^3} \\ &= \int \frac{I}{c}\frac{|d\mathbf{r}'|R\sin\frac{\pi}{2}}{R^3}\mathbf{e}_z \\ &= \int_0^{2\pi}\frac{I}{c}\frac{R^2 d\theta}{R^3}\mathbf{e}_z = \frac{2\pi I}{cR}\mathbf{e}_z,\end{aligned}$$

which matches the previous large-n limit in Q12.

V. Lorentz force on a wire in a uniform magnetic field (HW3 Q14)

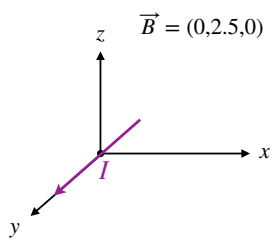

Figure 5: (a) A straight wire

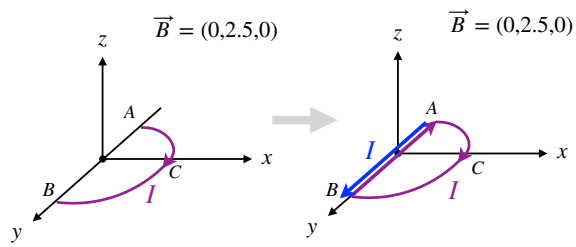

Figure 6: (b) A half-circle wire is equivalent to a current loop plus a straight wire

(a) By definition, the Lorentz force on this straight wire is given by

$$\begin{aligned}
\mathbf{F} &= \int \frac{I}{c} d\mathbf{r} \times \mathbf{B}(\mathbf{r}) \\
&= \int_{-1}^{1} \frac{I}{c} dx \begin{vmatrix} \mathbf{e}_x & \mathbf{e}_y & \mathbf{e}_z \\ 1 & 0 & 0 \\ 0 & 2.5 & 0 \end{vmatrix} \\
&= \frac{5I}{c} \mathbf{e}_z
\end{aligned}$$

(b) Instead of blindly calculating the Lorentz force, we use a trick here without any calculation. Notice that a half-circle wire is equivalent to a current loop plus a straight wire with directions shown in Fig.(6)(Why?). Based on Q1, we know that the Lorentz force on a current loop vanishes in a uniform magnetic field. Therefore, the force on the half-circle wire loop is just the force on the blue straight wire, which is just the force in (a) $\mathbf{F} = \frac{5I}{c}\mathbf{e}_z$.

VI. Poof of $\oint_c d\mathbf{r} = 0$ when the path is a circle (HW3 Q18)

1. **Cartesian coordinates**
   The parametric equation of circle in Cartesian coordinates is $\mathbf{r}_c = (R\cos\theta, R\sin\theta),\ \theta \in [0, 2\pi)$. Therefore,

$$\oint_c d\mathbf{r} = \int_0^{2\pi} \frac{d\mathbf{r}_c}{d\theta} d\theta = \int_0^{2\pi} (-R\sin\theta, R\cos\theta) d\theta = \mathbf{0}$$

2. **Polar coordinates**
   The parametric equation of circle polar coordinates is $r = R,\ \theta \in [0, 2\pi)$, assuming the origin is at the center of the circle. Therefore,

$$\begin{aligned}
\oint_c d\mathbf{r} &= \oint_c dr\mathbf{e}_r + \oint_c r d\theta \mathbf{e}_\theta \\
&= \mathbf{0} + R \oint_c d\theta \mathbf{e}_\theta
\end{aligned}$$

However, in the second term the unit vector $\mathbf{e}_\theta$ is NOT a constant vector but a function of angle θ, i.e. $\mathbf{e}_\theta(\theta)$. Therefore, $\mathbf{e}_\theta(\theta)$ can not be pulled out of the integral,

$$\boxed{\oint_c d\theta \mathbf{e}_\theta \neq \mathbf{e}_\theta \oint_c d\theta}$$

To calculate the second term $\oint_c d\theta \mathbf{e}_\theta$, we resort to Cartesian coordinates, in which $\mathbf{e}_\theta = -\sin\theta \mathbf{e}_x + \cos\theta \mathbf{e}_y$.

$$\oint_c d\theta \mathbf{e}_\theta = \int_0^{2\pi} d\theta(-\sin\theta \mathbf{e}_x + \cos\theta \mathbf{e}_y) = \mathbf{0}$$

In the equation above, $\mathbf{e}_x$ and $\mathbf{e}_y$ are constant vectors, so they can be safely pulled out of the integral.